Solving Differential Equation Descriptions: A Worked Example

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Find the output of the system described by the differential equation:

$$\frac{d^2}{dt^2}y(t) + 3\frac{d}{dt}y(t) + 2y(t) = 10\,x(t)$$

if $x(t) = \cos(t)\,u(t)$, $y(0^-) = -4$, $\left.\frac{d}{dt}y(t)\right|_{t=0^-} = 3$

Solution –

1) Steady-state response $y_s(t)$

a) Write steady-state form of $x(t)$ as $\sum_{p=1}^{Q} \alpha_p e^{s_p t}$

$$\cos(t) = \frac{1}{2}e^{jt} + \frac{1}{2}e^{-jt} \Rightarrow \alpha_1 = \alpha_2 = 1/2,\ s_1 = j,\ s_2 = -j$$

b) Find $H(s) = \dfrac{\sum_{k=0}^{M} b_k s^k}{\sum_{k=0}^{N} a_k s^k} = \dfrac{10}{s^2 + 3s + 2}$

$b_0 = 10$
$a_0 = 2$
$a_1 = 3$
$a_2 = 1$

C) $y_s(t) = \alpha_1 H(s_1) e^{s_1 t} + \alpha_2 H(s_2) e^{s_2 t}$

$= \frac{1}{2} H(j) e^{jt} + \frac{1}{2} H(-j) e^{-jt}$

$H(s) = \frac{10}{s^2+3s+2}$

$H(j) = \frac{10}{-1+3j+2} = \frac{10}{1+3j} = \frac{10(1-3j)}{10} = 1-3j$

$H(-j) = \frac{10}{-1-3j+2} = 1+3j$

$y_s(t) = \frac{1}{2}\left[1 \cdot e^{jt} - 3j\, e^{jt} + 1 \cdot e^{-jt} + 3j\, e^{-jt}\right]$

$= \frac{e^{jt} + e^{-jt}}{2} + 3\left(\frac{e^{jt} - e^{-jt}}{2j}\right)$

$y_s(t) = \cos(t) + 3\sin(t)$

2) Transient Response $y_t(t)$

$$y_t(t) = \sum_{\ell=1}^{N} c_\ell e^{\tilde{s}_\ell t}$$

$\tilde{s}_\ell$ roots of $a_2 s^2 + a_1 s + a_0 = 0$

$$s^2 + 3s + 2 = 0$$

Factor: $(s+2)(s+1) = 0 \Rightarrow \tilde{s}_1 = -2, \tilde{s}_2 = -1$

$$y_t(t) = c_1 e^{-2t} + c_2 e^{-t}$$

Note: $\tilde{s}_\ell$ are distinct from $s_1 = j, s_2 = -j$

Solution: $y(t) = y_s(t) + y_t(t)$

$$y(t) = \cos(t) + 3\sin(t) + c_1 e^{-2t} + c_2 e^{-t}, \quad t > 0$$

Find c_1, c_2 using initial conditions: $y(0^-) = -4$, $\frac{d}{dt}y(t)\Big|_{t=0^-} = 3$

$$y(t) = \cos(t) + 3\sin(t) + c_1 e^{-2t} + c_2 e^{-t}, \quad t > 0$$

a) RHS $10x(t)$ for $x(t) = \cos(t)\,u(t)$ no impulses (or derivatives) at $t=0$

$$y(0^+) = -4, \quad \frac{d}{dt}y(t)\Big|_{t=0^+} = 3$$

b) Write equations for c_1, c_2

$$-4 = \cos(0) + 3\sin(0) + c_1 + c_2 \quad \Rightarrow \quad -5 = c_1 + c_2$$
$$3 = -\sin(0) + 3\cos(0) - 2c_1 - c_2 \qquad 0 = -2c_1 - c_2$$

so $c_1 = 5$, $c_2 = -10$

$$y(t) = \cos(t) + 3\sin(t) + 5e^{-2t} - 10e^{-t}, \quad t > 0$$